

Machine Learning and Simulations

-- Advanced Technology, Modeling and Simulation
for Networking

Overview

- Computer Networks (done)
- Programming (done)
- Network Simulation (done)
- Advanced Techniques and Tools

Machine Learning for Simulations

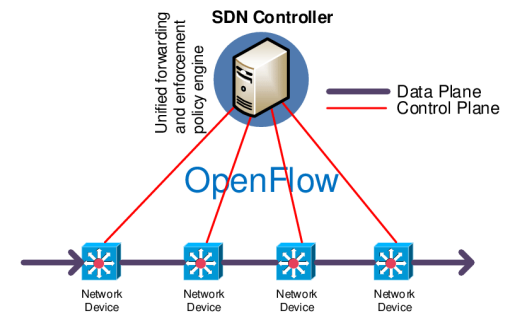
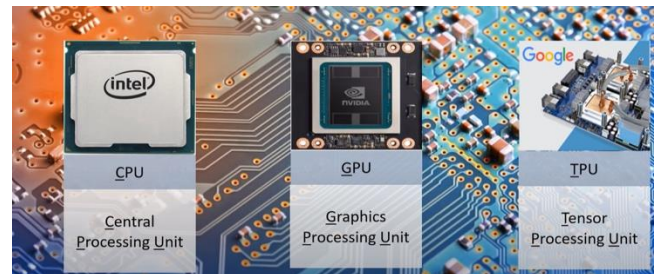
Why Machine Learning?

- Motivation

- **System complexity**: # of access points and users, dynamics
(hard to model analytically)
- **Space of algorithm** becomes very large
(complex search for suitable algorithm)
- **Model/Algorithm deficiencies** due to imperfect information
(learn and adapt)

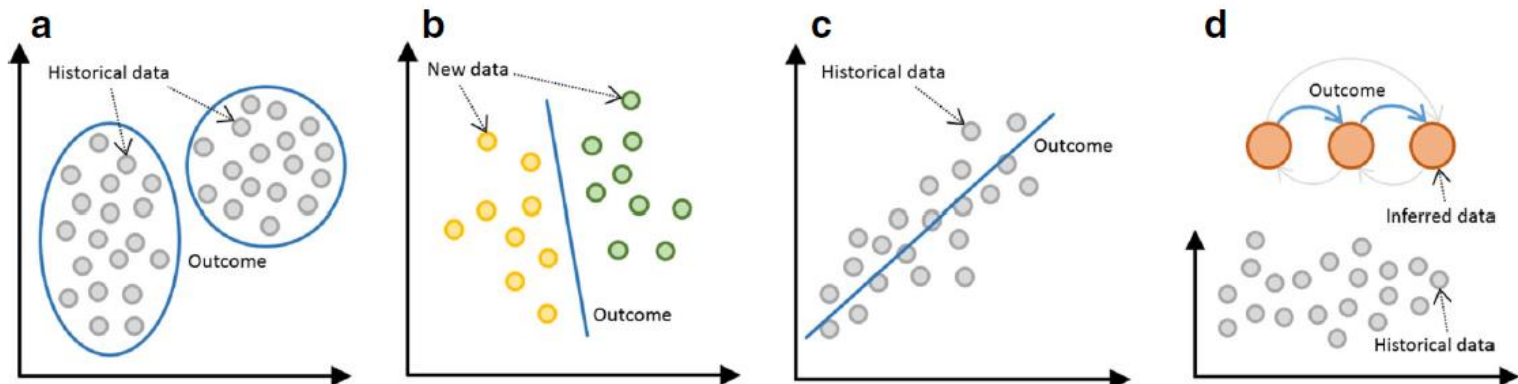
Why Machine Learning?

- Aligned with:
 - Big data – availability of large data sets
 - Fast computing (GPU, TPU)
 - Key technological advances in networking (NFV, SDN)

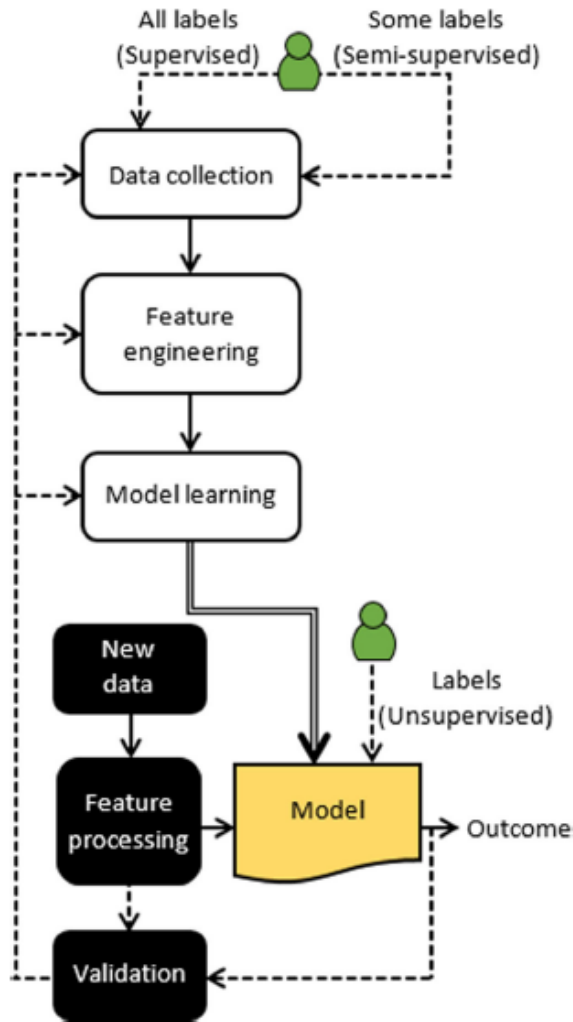


Machine Learning (ML)

- broad categories of problems that can leverage ML:
 - Clustering
 - Classification
 - Regression
 - Rule extraction



Machine Learning paradigms

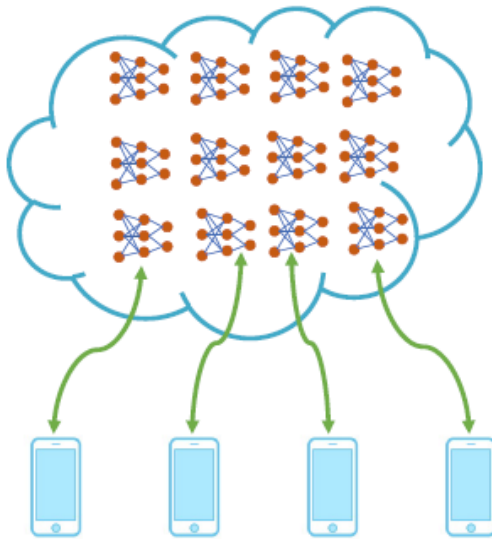


Machine Learning in Computer Networking

- Dataset collection
- Network traffic prediction
- Network routing
- Congestion control
- Resource management
- Network and data security

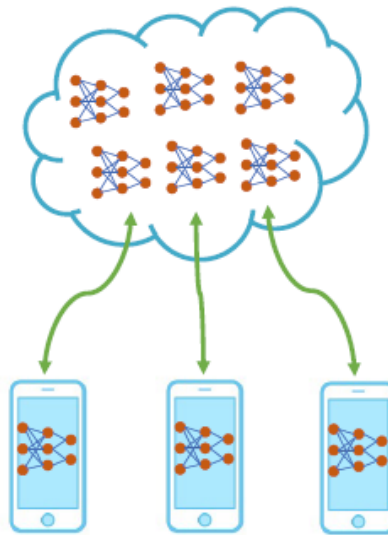
Networked Machine Learning Implementation

"Standard" ML



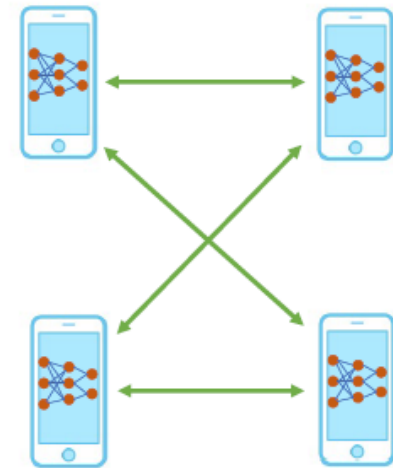
- ML in the cloud with dumb end-user devices
- All data is in the cloud
- Inference and decision making in the cloud
- No data privacy

Federated ML



- ML in the cloud + on-user-device ML
- Only part of the data is in the cloud
- Use the cloud but smartly
- Privacy-promoting

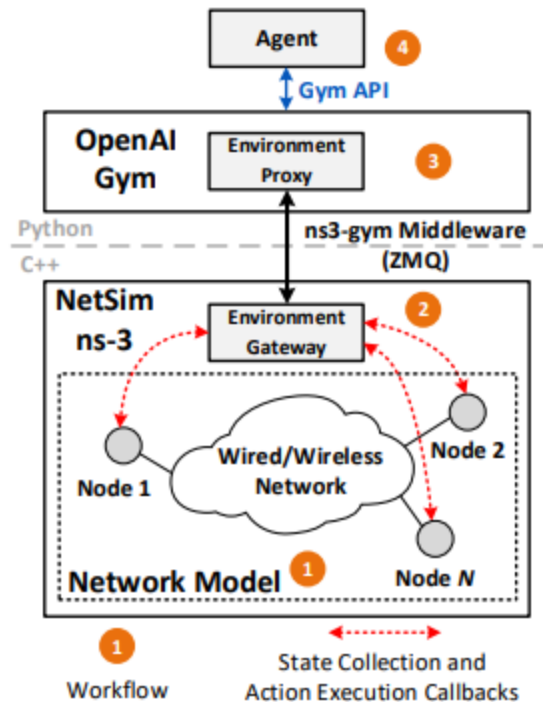
Decentralized ML



- No infrastructure (e.g., cloud) needed
- Data is fully distributed
- Collaborative intelligence
- Privacy-promoting (sharing models instead of data)

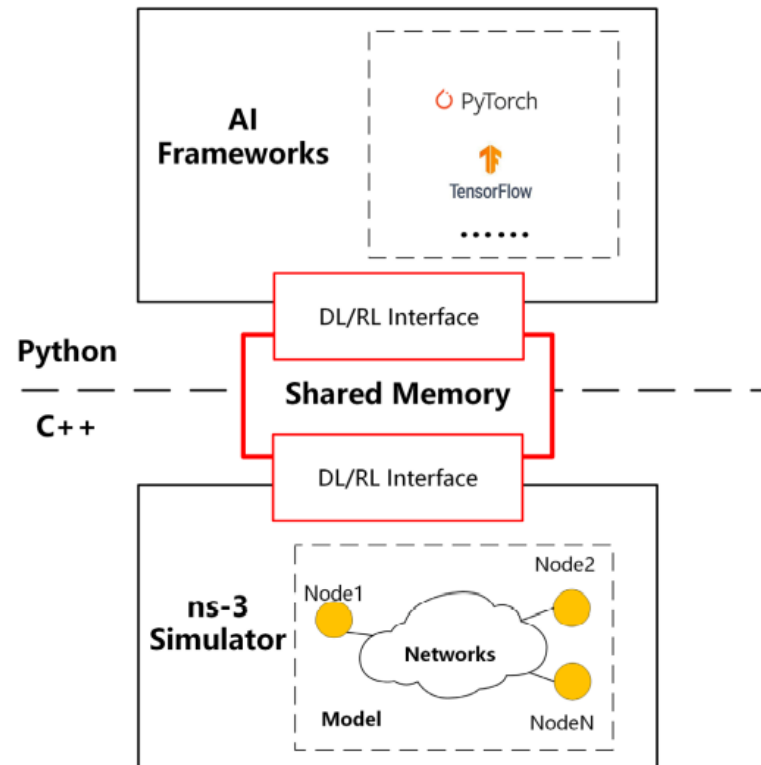
ns-3 AI/ML

- ns3-gym
 - The Playground for Reinforcement Learning in Simulation
 - <https://apps.nsnam.org/app/ns3-gym/>



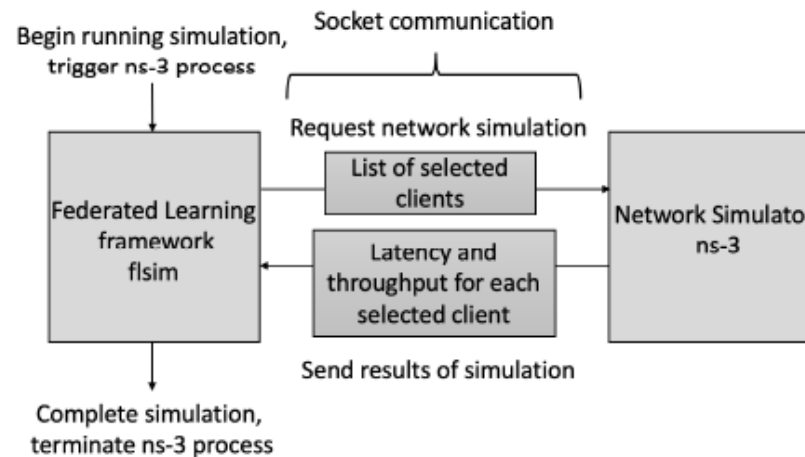
ns-3 AI/ML

- ns3-ai
 - connect ns-3 with AI/ML frameworks
 - <https://apps.nsnam.org/app/ns3-ai/>



ns-3 AI/ML

- ns3-fl
 - connect an existing FL simulator *flsim* with *ns-3*
 - <https://github.com/eeaireb/ns3-fl>



Open-Source Dataset

- RFDATA Factory
 - <https://www.rfdatafactory.com/datasets>
- OpenRAN Gym
 - <https://openrangym.com/datasets>
- 5G End-to-End Emulation
 - <https://github.com/cedric-cnam/5G3E-dataset>

Presentation

- Find your group (3)
- Prepare couple of slides
- Upload your slides (**with your group name**) by 9am Friday to:

https://drive.google.com/drive/folders/1cZzCqx41jt1wk0qFYyzZ108xl-VYauDG?usp=drive_link

MATLAB ToolBox

[WLAN Toolbox - MATLAB \(mathworks.com\)](#)